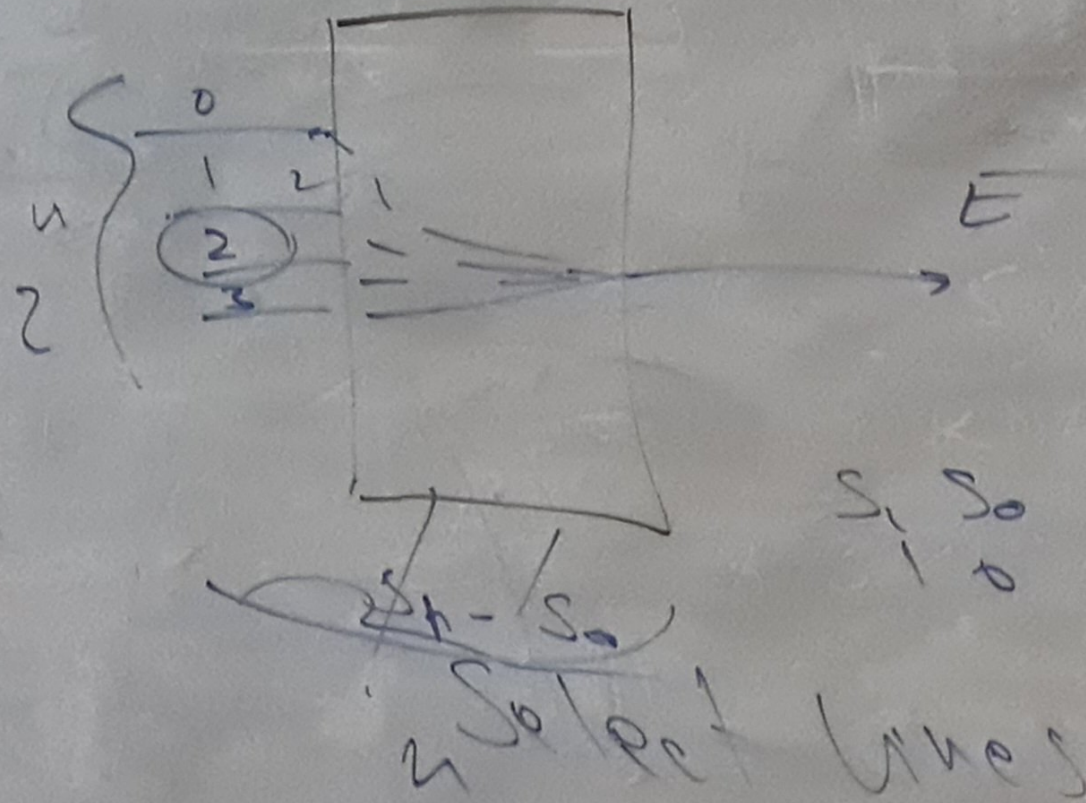
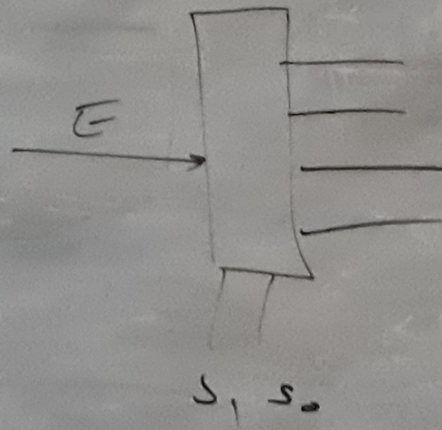


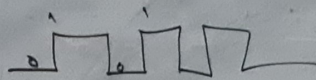
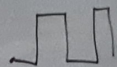
Multiplexer

②

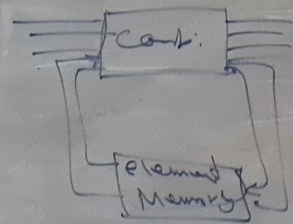


Demultiplexer -



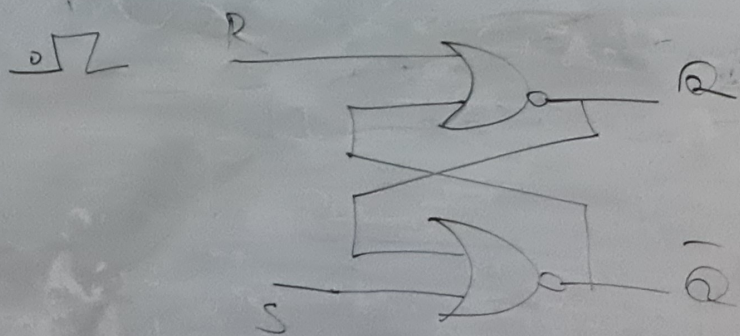


Sequential Circuit

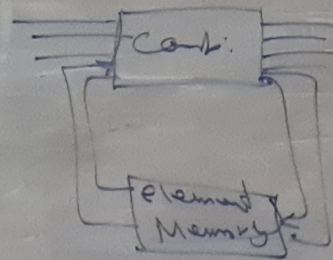


- 1. Synchronous — (clocked)
- 2. Asynchronous (N —)

SR-NOR (FF)



Sequential Circuit

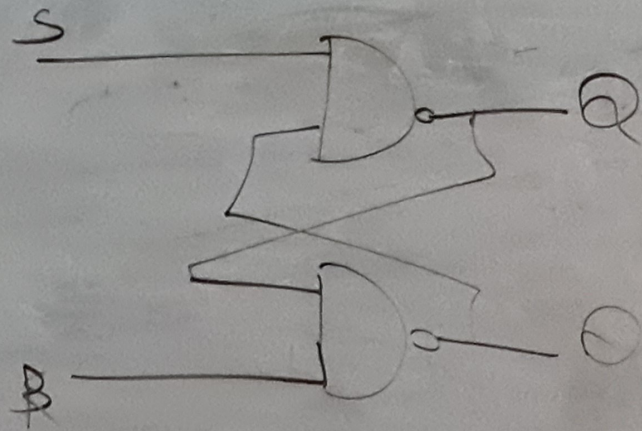
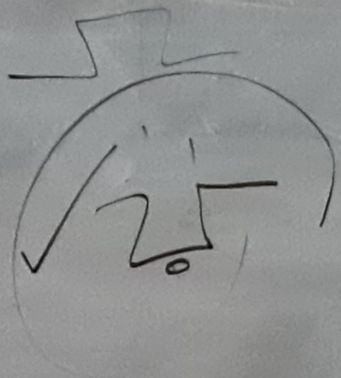


1. Synchronous (clocked)
2. Asynchronous (N —)

Sequential Circuit

S R	Q	$\bar{Q}$
1 0		
0 0		
0 1		
1 1		

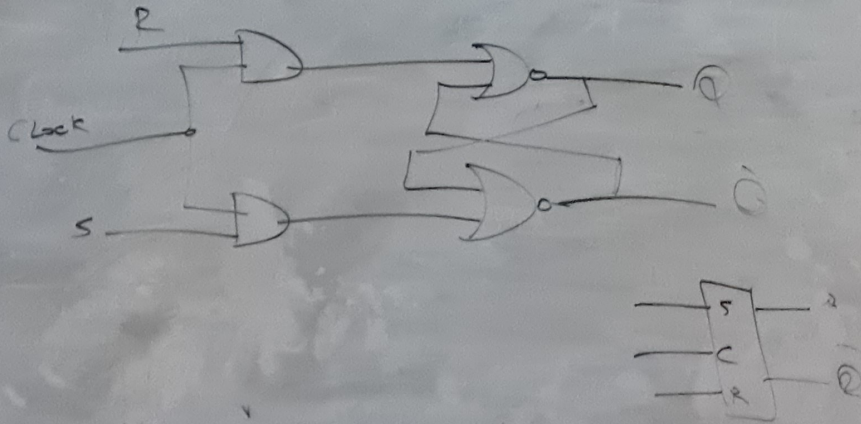
SR-NAND (FF)



Sequential Circuit

S	R	Q	Q ⁻
1	0	0	1
0	0	0	1
0	1	1	0
0	0	0	0
0	0		

RS-FF

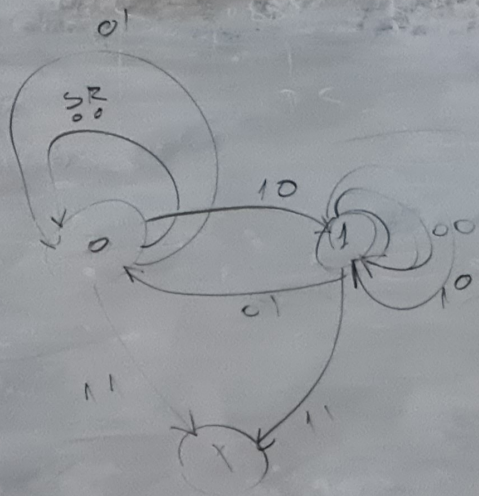


$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Locked FF

	$\bar{S}\bar{R}$	$\bar{S}R$	SR	$S\bar{R}$
$\bar{Q}(t)$			X	1
$Q(t)$	1	X	X	1

$$Q(t+1) = S + Q(t)\bar{R}$$



2. State Diagram

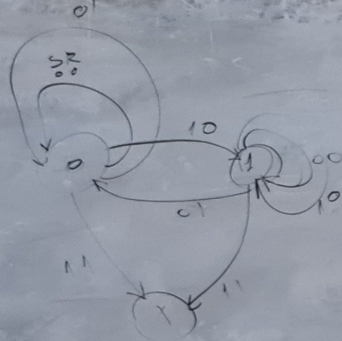
1. Truth Table

$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Locked FF

	$\bar{S}\bar{R}$	$\bar{S}R$	SR	$S\bar{R}$
$Q(t)$			X	1
$Q(t)$	1	X	X	1

$$Q(t+1) = S + Q(t)\bar{R}$$



2. State Diagram

1. Truth Table

$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Clock = F

	$\bar{S}\bar{R}$	$\bar{S}R$	SR	$S\bar{R}$
$\bar{Q}(t)$			X	1
$Q(t)$	1	X	X	1

$$Q(t+1) = S + Q(t)\bar{R}$$

